

## Lighting

### Colors

- Material color acts as per-channel reflection coefficient, not as light emitted by the object.

Each RGB value represents the fraction of incoming red, green, and blue light that the surface reflects.

- $$\text{Reflected Color} = \text{Light Color} \times \text{Material Color}$$

- Multiplication models light reflection: if a material reflects 50% of green light, then  
$$G_{out} = G_{in} \times 0.5$$

→ same applies ~~for~~ independently to the red & blue channels

### Light Sources

#### Ambient Light

- The ambient light present at certain location is the low-intensity light that arises from the many reflections of light on all nearby surfaces in an environment.
- Ambient light appears to come from every direction and thus, illuminates every part of an object uniformly.

#### Directional Light Source

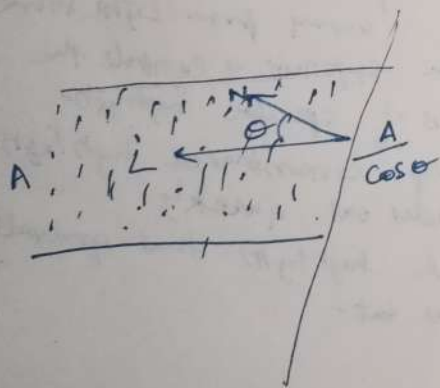
- Also known as an infinite light source,
- one that radiates light in a single direction from infinitely far away. Typically used to model light sources such as the sun, these have infinite range and intensity does not diminish over distance.



## Diffuse Reflection

~~Ambient~~ Ambient lighting by itself doesn't produce the most interesting results, but diffuse lighting however will start to give a significant visual impact on the object.

- Diffuse Lighting gives the object more brightness the closer its ~~face~~ frames are aligned to the light rays from a light source.



- Based on the figure the surface area illuminated by a beam of light increases as the angle  $\theta$  between the surface normal and direction of the light decreases, decreasing the intensity of incident light per unit area.

$$K_{\text{diffuse}} = DA + D \sum_{i=1}^n C_i \max\{N \cdot L_i, 0\}$$

$K \rightarrow$  Light Reflected toward the viewer from  $\theta$

$C_i \rightarrow$  Intensity of each of the  $n$  light sources.

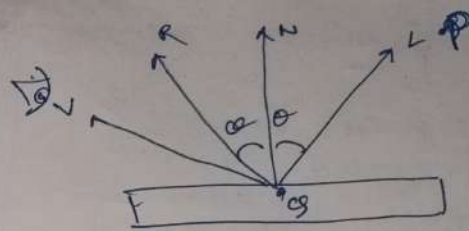
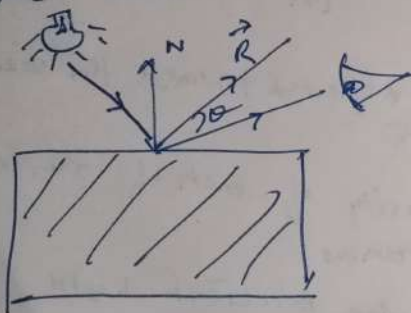
$A \rightarrow$  ~~Ambience~~ Ambient light intensity

$D \rightarrow$  Surface diffuse reflection color.



## Specular Lighting

- 9 similar to diffuse, specular lighting is based on the light's direction vector and the object's normal vectors, but the time it is also based on the view direction.



Intensity is given By:

$$S.C. \max \{R \cdot V, 0\}^m \quad (N.L70)$$

$S$  → Surface's specular reflection color.

$C$  → Intensity of incident light

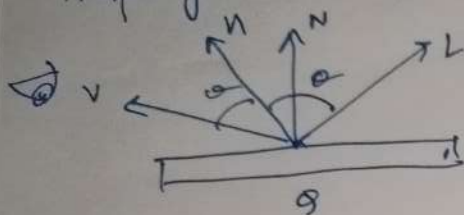
$(N.L70)$  → Boolean expression  
 Represents from showing  
 points in a surface that face  
 away from light source.

$m$  → specular exponent → controls the  
 sharpness of specular highlights

$m \uparrow$  → sharp concentrated highlight  
 & fades out quickly

$m \downarrow$  → dull highlight that gradually  
 fades out.

halfway vector Alternative



$$K_{\text{specular}} = S \sum_{i=1}^m \left( \frac{1}{2} \max \{ \mathbf{n} \cdot \mathbf{h}_i, 0 \}^m \right) \quad (\text{L. L'970})$$

$$\mathbf{h}_i = \frac{\mathbf{L}_i + \mathbf{V}_i}{|\mathbf{L}_i + \mathbf{V}_i|}$$